Byteclub

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Domain : Social Impact & Sustainability

NOTE: Do check out the **last slide** for **demo link** and **github link to get hands-on experience of our solution**.Hardware implementation is also added

Problem Statement

PS 3.2 | Lightweight Telemedicine Prototype for Low-Bandwidth Coastal & Hill Talukas

Context & Challenge

- Rural and underserved communities in India face a persistent gap in accessible, quality healthcare.
- Patients travel long distances to reach under-resourced PHCs, specialists are scarce, and unreliable connectivity makes digital tools fail in the field.
- ASHA workers(frontline healthcare worker) lack triage support, records remain paper-based and fragmented, and referrals ignore hospital capacity, causing delays and futile transfers.
- Emergencies are escalated verbally, follow-up for high-risk mothers and chronic patients is weak, and disease outbreaks are detected late because reporting is manual.
- The challenge is to build an offline-capable, multilingual, interoperable system that connects villages, PHCs and district hospitals, enabling timely, continuous, proactive care.

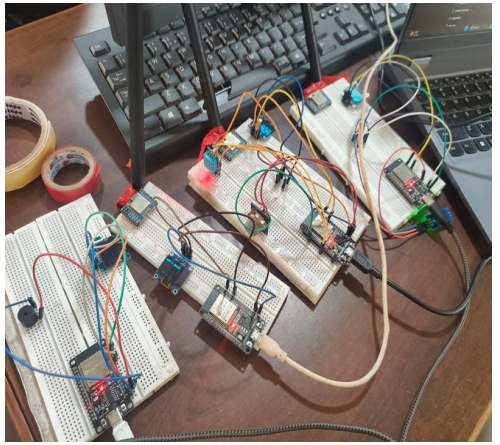




Solution Overview

PROPOSED SOLUTION

- **SeVa-Setu is an offline-first care-continuity framework, built for rural communities where no network exists.**
- **Our solution connects patients, ASHA workers and PHCs through on-device AI triage, and a dedicated LoRa network that reaches patients even in no/low connectivity areas.**
- **It acts as a layer on top of existing public health infrastructure, not a replacement . Every encounter quietly feeds into our predictive dashboard and flags a potential outbreak**



LORA Relay module



Wearable (Smart Health Monitoring)

****PLEASE GO THROUGH THE GITHUB FOR DETAILED OVERVIEW** CLICK HERE**

UNIQUE SELLING POINTS

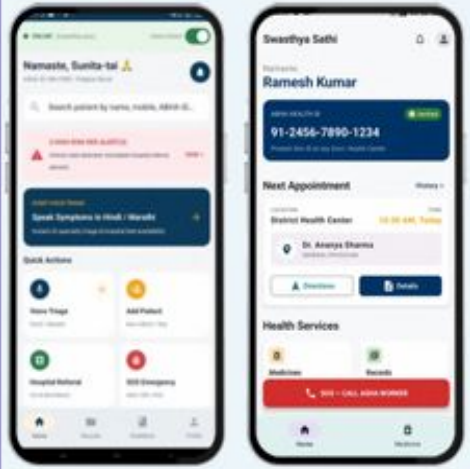
True dead-zone LoRa bridge	**108/IHIP integrated**	**Free outbreak detection**
Micro-containment zone alerts	**Works in full to zero signal signal**	**ABHA-linked, FHIR**

Technical Approach

1 FIELD APPS

ASHA WORKER APP

PATIENT APP



ASHA VISIT (Frontline capture)

PATIENT (Self Report)

VOICE + MULTILINGUAL

VITALS

SOS/SMS/IVR (For Patients)

2 ON DEVICE AI

VOICE/ASR

NLP (multilingual)

RULE+ML TRIAGE

EXPLAINABLE RISK

G Y R

SNOMED/ICD

PROTOCOL-BOUND ASSISTANT

3 RESILENT SYNC

SQLLite (On ASHA Worker app)

PRIORITY QUEUE (Encrypted)

RED YELLOW GREEN

Online (Cloud)

Lora Network

PHC Gateway

VitalPack (Compressed code)

YELLOW (Doctor reply)

RED (108 dispatch)

Offline (Locally stored)

Online (Sync to cloud)

4 AWS BACKEND

CLOUD ARCHITECTURE(AWS)

SERVICES(ECS/FARGATE)

- PATIENT
- TRIAGE
- REFERRAL
- APPOINTMENT
- INVENTORY
- FOLLOWUP OUTBREAK

EVENTS AND NOTIFICATIONS

- SQS
- SNS
- EVENTBRIDGE

SECURITY

- COGNITO (AUTH)
- IAM/RBAC
- KMS (ENCRYPTION)
- CLOUD WATCH (MONITORING)

INTEGRATIONS (WITH GOVT.)

- 108/112 (EMERGENCY)
- e-SANJEEVANI(TELECONSULT)
- ABHA/ABDM (HEALTH ID)
- FHIR R4 (STANDARD)

5 DATA+ DASHBOARD

DATA LAYER

RDS (PostgreSQL)

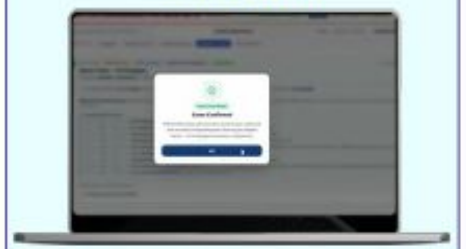
S3 (Downloads)

FHIR R4| SNOMED CT|LONC | ICT 10

PHC DASHBOARD- (ADMIN/DOCTOR)

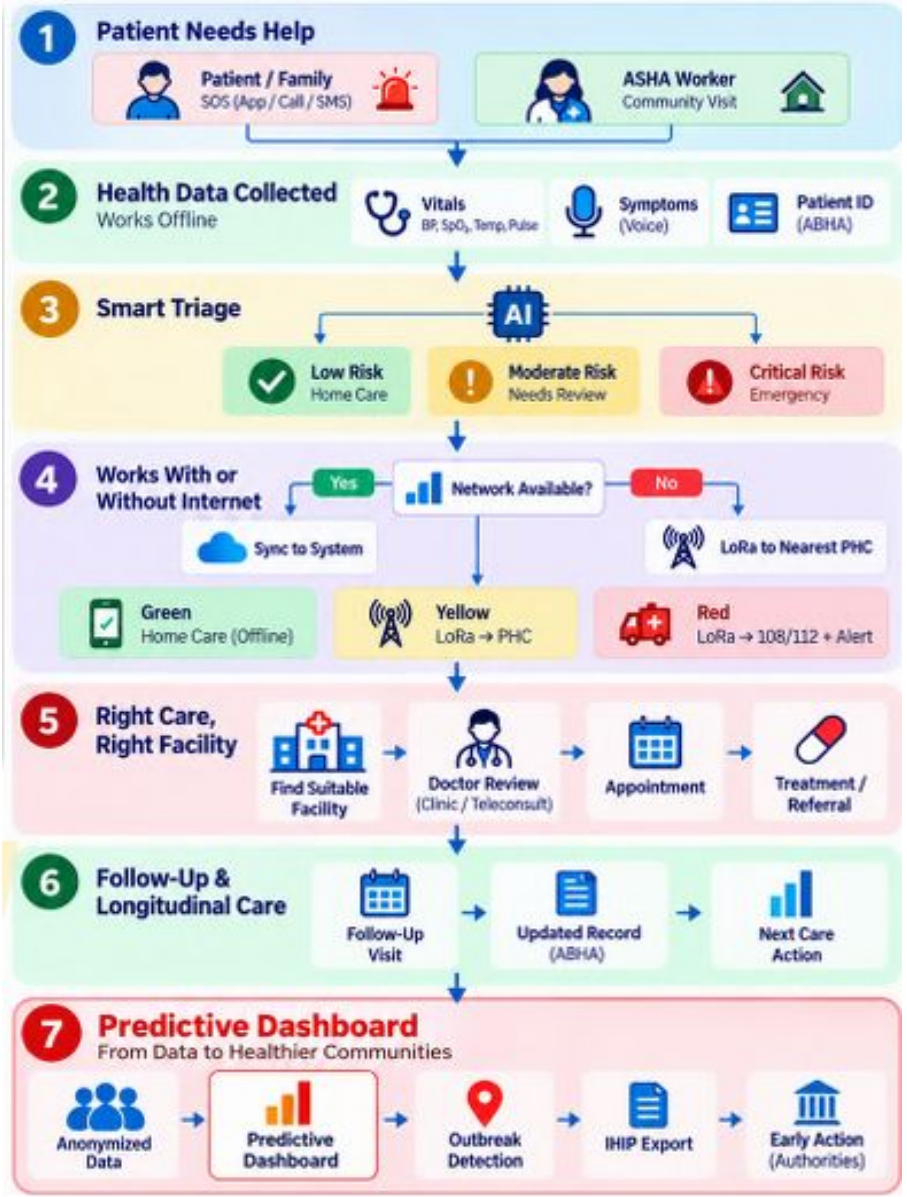


DISTRICT PREDICTIVE/OUTBREAK DASHBOARD



CORE TECH STACK

Diagrams/Flow Charts/Visuals



WORKFLOW

- **Entry:** A case starts one of two ways, a patient raises their own SOS, or an ASHA worker runs a routine visit both feed into the exact same engine from here on
- **Capture:** BLE sensor kit records vitals, voice captures symptoms in the local language no internet needed for this step
- **Triage:** On device engine instantly classifies the case as Green, Yellow, or Red, with a plain language reason attached never a bare score
- **Escalation path:** If there's zero cellular signal, a Yellow or Red case relays over a dedicated LoRa radio bridge straight to the nearest PHC
- **Resolution:** The case reaches a PHC/hospital that can actually treat it closing the loop from first contact to care, without ever depending on continuous connectivity

Impacts and Benefits

SOCIAL IMPACT

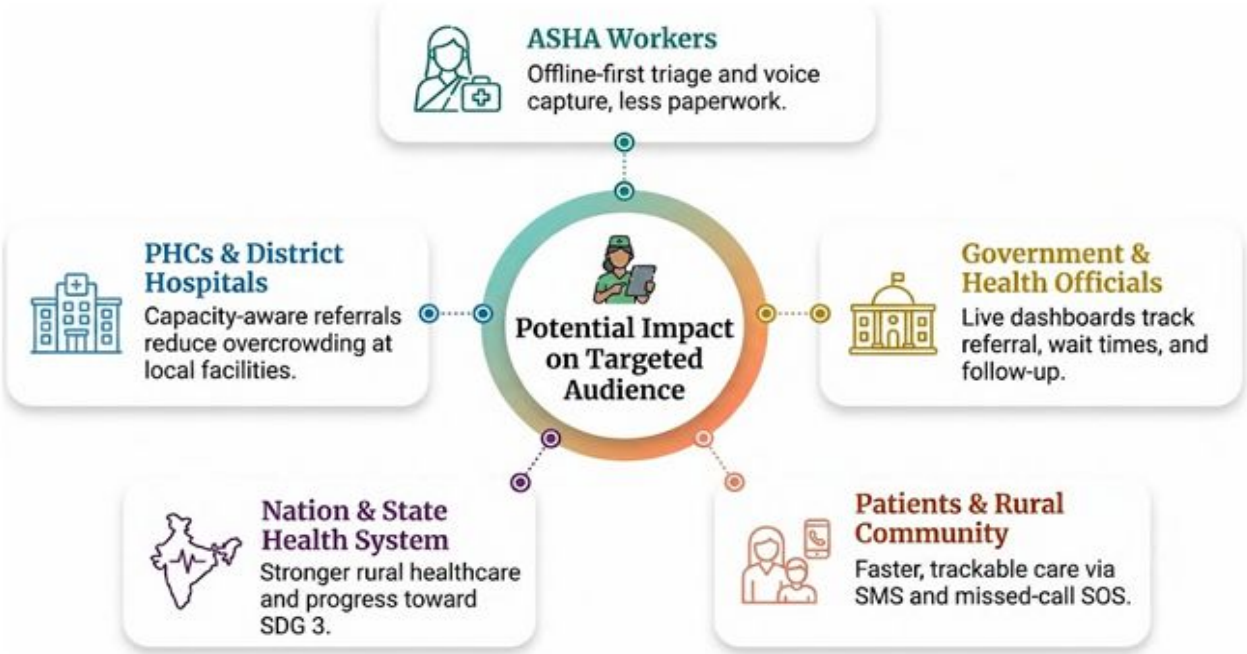
- Rural households travel 5–25 km to reach the nearest PHC,our SevaSetu cuts unnecessary trips.
- Explainable rule based risk scoring builds trust where every flag comes with a doctor-readable reason,not just a black box.
- Voice first triage in all Indian languages, along with SMS/SOS fallback.
- Reaches true cellular dead zones via a dedicated LoRa network ,not just SMS/SOS.(Works in low/no network regions)

ECONOMIC IMPACT

- 49% of Indian households face devastating loss i.e it costs a life from delayed/hospitalized care, that can be prevented
- 16% of households borrow or sell assets to cover avoidable healthcare costs; now it can be reduced.
- Using pay-as-you-go cloud infrastructure instead of expensive hardware servers means the government only pays for what it actually uses.

HEALTH SYSTEM

- 75% of India's health infrastructure serves only 27% of the population (this platform extends real reach into the underserved 73% rural population)
- Strengthens, not replaces, the public system plugs directly into ABDM/ABHA and eSanjeevani.
- Automated follow-up alerts,close the maternal/child care continuity gap that today simply goes untracked.



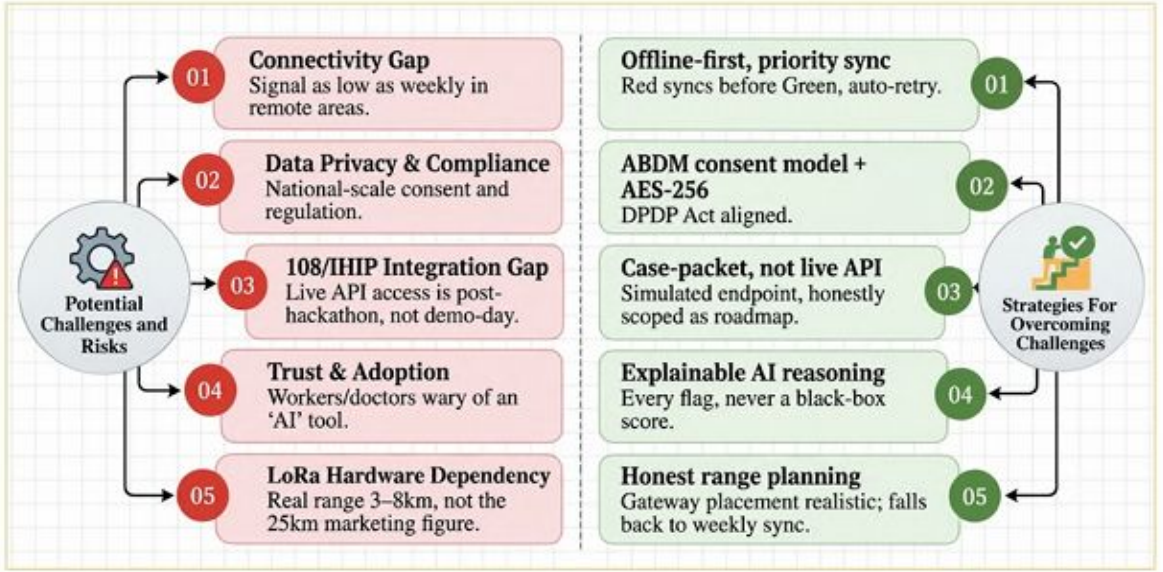
WHY SEVASETU?

Feature / Aspect	SevaSetu (Our App)	eSanjeevani	CommCare-based ASHA Apps
Fully Offline AI Triage	✓	✗	✗
True Dead-Zone LoRa Relay	✓	✗	✗
Explainable Risk Score	✓	✗	✗
Capacity & Stock-Aware Referral	✓	✗	✗
ABHA/FHIR-Native Records	✓	✓	✗
Voice-First Multilingual NLP	✓	✗	✗

Business Impact & Scalability

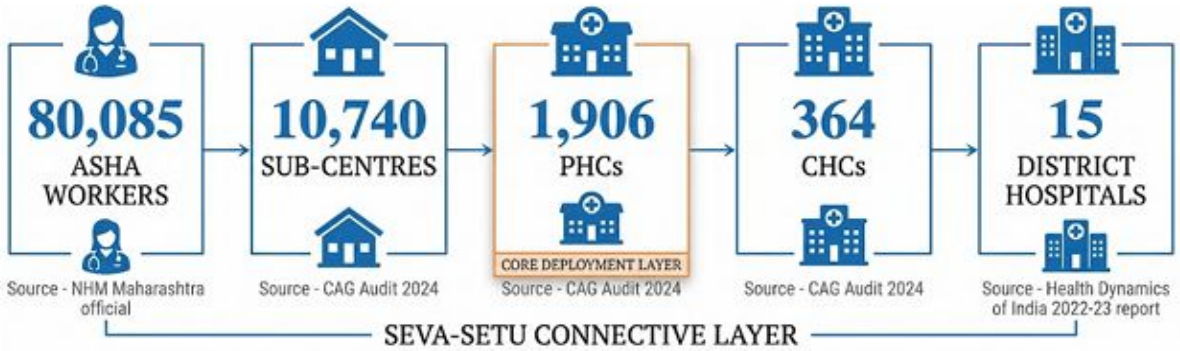
FEASIBILITY

- **Technical Stack:** Native Kotlin app built in Android Studio no cross platform runtime overhead, runs on the low-end Android devices
- **On-Device AI:** Hybrid rule engine, explainable scoring with zero network dependency.
- **Data Compliance:** Data model designed for ABDM / FHIR R4, SNOMED CT, LOINC, ICD-10 alignment from day one.
- **Security:** AES-256 local encryption + HMAC-SHA256 payload signing + TLS in transit, stated explicitly, not assumed.
- **Connectivity Layer:** LoRa network bridges true cellular dead zones, no telecom licensing needed, realistic 3–8km range.
- **Operational:** True offline-first store with priority ordered sync (Red > Yellow > Green), validated against real rural constraints down to weekly-connectivity villages.



VIABILITY

- **Market:** Direct pipeline through Government of Maharashtra's own SIH mandate; taps a national base of 70+ crore ABHA-linked patients already live under ABDM.
- **Policy & Revenue:** Fits Ayushman Bharat Digital Mission (ABDM), National Health Mission, and eSanjeevani integration , positions for state NHM budget allocation and per-PHC licensing, not a cold-start product.
- **Investment & ROI:** Managed AWS stack (ECS Fargate, RDS, SQS) keeps infrastructure cost pay-as-you-go.
- **Scalability:** Stateless microservices on ECS Fargate onboard new PHCs and ASHA workers without redesign.
- **Retention:** Explainable AI triage builds doctor trust; patient app for smartphone users, SMS / missed-call IVR for everyone else full reach regardless of device ownership.



80K+ FRONTLINE WORKERS → 1,906 PHCs → 364 CHCIDE REFERRAL NETWORK

Additional links

- **GitHub Repo link:** *https://github.com/Yokesveren/SevaSethu-Team_Cantina-*
- **Demo Video Link:** https://youtu.be/ap-gmuTtrpQ
- **Animated Video Link:** https://youtu.be/CpFEbZ00S2Y

Please go through all the links for better clarity

RESEARCH AND REFERENCES

- **LoRa spectrum & range validation:** https://zbotic.in/lora-technology-guide-india-long-range-iot-without-wifi/
- **eSanjeevani national telemedicine platform :** 43 crore teleconsultations as of Nov 2025, fully ABHA/ABDM-integrated https://esanjeevani.mohfw.gov.in/
- **Ayushman Bharat Digital Mission (ABDM) :** India's official digital health backbone; 73.98 crore ABHA accounts created, validating your ABHA-anchored record design. https://abdm.gov.in/
- **IMNCI Facilitator Guide 2023, MoHFW:** the government protocol your rule-engine's red-flag thresholds trace back to. https://nhm.gov.in/images/pdf/programmes/child-health/guidelines/IMNCI-Module-2023-For-Medical-Officers/IMNCI-Facilitator-Guide-Medical-Officer-2023.pdf
- **HL7 FHIR R4 specification:** the resource model your VitalPack layer converts to on sync. https://hl7.org/fhir/R4/
- **Rural CHC specialist shortage report, Union Health Ministry:** the 70% vacancy gap your referral engine solves. https://www.chronicleindia.in/current-affairs/12472-rural-health-centres-face-70-specialist-doctor-shortage

DATASETS USED

- **India Disease & Outbreak Datasets (aggregator):** https://dataful.in/datasets/?q=disease
- **IndicConformer :** Offline multilingual ASR https://github.com/AI4Bharat/IndicConformerASR
- **NFHS-5:** Population, health & nutrition indicators https://www.data.gov.in/resource/
- **HMIS:** Facility & health-system service data https://www.data.gov.in/catalog/item-wise-monthly

Field Research & Validation



A GMeet interview with Mrs. Jaya, a nurs working in a rural district of Karbataka, understand frontline healthcare challenge and existing workflows.